ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

MATHEMATICS

PRACTICE EXAM 3 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 8 Mathematics practice exam 3.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct answers.

QUESTION 1

1 1.1 C ✓ 1.2 A ✓ 1.3 D ✓ 1.4 B ✓ 1.5 D ✓ 1.6 B ✓ 1.7 C ✓ 1.8 A ✓ 1.9 B ✓ 1.10 D ✓ [10]

QUESTION 2

2.1.1 $24 \times \text{R}350 = \text{R}8\,400$ ✓ (1)

2.1.2 $\text{R}4\,800 + 24 \times \text{R}160$ ✓ $= \text{R}4\,800 + \text{R}3\,840 = \text{R}8\,640$ ✓ (2)

2.1.3 Deal A ✓ is cheaper by $\text{R}8\,640 - \text{R}8\,400 = \text{R}240$ ✓ (2)

2.2.1 $15\% \times \text{R}9\,000 = \text{R}1\,350$ ✓ (1)

2.2.2 $\text{Balance} = \text{R}9\,000 - \text{R}1\,350 = \text{R}7\,650$ ✓; $\text{interest} = \text{R}7\,650 \times 0{,}12 \times 2 = \text{R}1\,836$ ✓ (2)

2.2.3 $(\text{R}7\,650 + \text{R}1\,836) \div 24$ ✓ $= \text{R}9\,486 \div 24 = \text{R}395{,}25$ ✓ (2)

2.2.4 $\text{R}1\,350 + \text{R}9\,486$ ✓ $= \text{R}10\,836$ ✓ (2)

[12]

QUESTION 3

3.1 (a) $27x^6 \times 2x$ ✓ $= 54x^7$ ✓ (6)
(b) $3a^3$ ✓ b^2 ✓
(c) $(-8)(-1) + 1$ ✓ $= 9$ ✓

3.2 (a) $x = 5$ ✓ (b) $x = 3$ ✓ (2)

3.3 $7 - (6 \times 2)$ ✓ $= -5\ ^\circ\text{C}$ ✓ (2)

[10]

QUESTION 4

4.1 $^3/_4 \times {}^{10}/_6$ ✓ $= {}^{30}/_{24}$ ✓ $= 1\ {}^1/_4$ cups ✓ (3)

4.2 (a) $^7/_5 \times {}^5/_2$ ✓ $= {}^7/_2 = 3\ {}^1/_2$ ✓ (5)
(b) $0{,}09 + 0{,}7$ ✓ $= 0{,}79$ ✓
(c) $87{,}5\%$ ✓

[8]

QUESTION 5

5.1 (a) $T_n = n^2$ ✓ (b) 144 ✓ (c) $n^2 = 400 \rightarrow n = 20$ (the 20th term) ✓ (3)

5.2 (a) $6a^2 - 8ab - a^2 + 8ab$ ✓✓ $= 5a^2$ ✓ (5)

(b) $3x^2 - 2x + 1$ ✓✓

5.3 (a) $6x - 3 = 4x + 8$ ✓ $\rightarrow 2x = 11$ ✓ $\rightarrow x = 5{,}5$ ✓ (6)

(b) Multiply by 6: $3x + 2x = 60$ ✓ $\rightarrow 5x = 60$ ✓ $\rightarrow x = 12$ ✓

5.4 $n + (n + 1) + (n + 2) = 87$ ✓ $\rightarrow 3n + 3 = 87 \rightarrow n = 28$ ✓ $\rightarrow$ 28, 29 and 30 ✓ (3)

5.5 (a) $8(12) + 15 = \text{R}111$ ✓ (3)

(b) $8d + 15 = 95 \rightarrow 8d = 80$ ✓ $\rightarrow d = 10$ km ✓

[20]

QUESTION 6

6.1 (a) $\angle\text{TUV} = 50°$ ✓ (alternate angles, PQ ∥ RS) ✓ (6)

(b) $\angle\text{TVU} = 70°$ ✓ (alternate angles, PQ ∥ RS) ✓

(c) $\angle\text{UTV} = 180° - 50° - 70° = 60°$ ✓ (adjacent angles on a straight line PQ, or sum of the angles of ΔTUV) ✓

6.2 Yes, $\Delta\text{ABC} \equiv \Delta\text{DEF}$ ✓ (SAS) ✓ (2)

6.3 $3y + 15° = 5y - 25°$ (opposite angles of a parallelogram are equal) ✓ $\rightarrow 2y = 40° \rightarrow y = 20°$ ✓ (4)

$\angle\text{A} = 3(20°) + 15° = 75°$ ✓ $\angle\text{B} = 180° - 75° = 105°$ (co-interior angles, AD ∥ BC) ✓

6.4 $\angle\text{X} + \angle\text{Y} = 120°$ (exterior angle of a triangle) ✓ $\rightarrow 2\angle\text{Y} + \angle\text{Y} = 120° \rightarrow \angle\text{Y} = 40°$ ✓ $\angle\text{X} = 80°$ ✓ (3)

6.5 $(6 - 2) \times 180° \div 6 = 120°$ ✓ (1)

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QUESTION 7

7.1 Diagonal $= \sqrt{(40^2 + 30^2)}$ ✓ $= 50$ m ✓; saving $= (40 + 30) - 50 = 20$ m ✓ (3)

7.2 (a) $h^2 = 10^2 - 6^2 = 64$ ✓ $\rightarrow h = 8$ cm ✓ (b) $\text{A} = ½ \times 12 \times 8 = 48\ \text{cm}^2$ ✓ (3)

7.3 (a) $\text{A} = {}^{22}/_{7} \times 3{,}5^2$ ✓ $= 38{,}5\ \text{m}^2$ ✓ (4)

(b) $\text{C} = 2 \times {}^{22}/_{7} \times 3{,}5$ ✓ $= 22$ m ✓

7.4 Area of one face $= 150 \div 6 = 25\ \text{cm}^2$ ✓ $\rightarrow$ side $= 5$ cm ✓ $\rightarrow \text{V} = 5^3$ ✓ $= 125\ \text{cm}^3$ ✓ (4)

[14]

QUESTION 8

8.1.1 $800\ \text{m}$ ✓ (1)

8.1.2 He stopped (rested / visited a shop) – the distance did not change ✓ (1)

8.1.3 $1\ 400\ \text{m}$ ✓ (1)

8.1.4 $800 \div 10$ ✓ $= 80\ \text{m/min}$ ✓ (2)

8.1.5 The first 10 minutes (800 m compared with 600 m; the graph is steeper) ✓ (1)

[6]

QUESTION 9

9.1 (a) ${}^{4}/_{8} = {}^{1}/_{2}$ ✓ (b) primes 2, 3, 5, 7: ${}^{4}/_{8} = {}^{1}/_{2}$ ✓ (c) multiples 3 and 6: ${}^{2}/_{8}$ ✓ $= {}^{1}/_{4}$ ✓ (4)

[4]

TOTAL: 100

END OF MARKING GUIDELINE